

# Forced or Frantic? Deleveraging Cascades and Price Dislocation on a Transparent Perpetual-Futures Venue

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Working paper, draft, June 2026

**Abstract.** Liquidation cascades are widely blamed for sharp price dislocations in crypto derivatives markets, but the public liquidation feeds most studies rely on are throttled and incomplete, and they cannot separate *forced* selling (margin liquidations) from *frantic but voluntary* position-closing. I exploit Hyperliquid, an on-chain perpetual-futures venue whose complete liquidation record is publicly available, to test whether forced-deleveraging events dislocate prices differently from voluntary churn. Using 162 liquidation-triggered events in BTC and ETH (August-December 2025), in raw event contrasts deleveraging events show larger peak dislocations, lower transitory shares (more persistent impact), and slower recovery than churn events. The hierarchy of these findings under scrutiny is the paper’s main message. The larger-dislocation result is the strong claim: it survives controlling for event magnitude (about 0.55 percentage points,  $p = 0.0001$ ), holds across trigger thresholds, both assets, and exclusion of the October-2025 cascade, and is consistent with an external funding-rate check, indicating the *mechanism*, not merely the size of the flush, matters. The greater persistence is a weaker claim, largely mediated by magnitude; and the slower recovery is only suggestive, its independence from size unresolved at this sample. The entire study uses only free, fully auditable data.

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## 1. Introduction

When leveraged positions are force-closed during a sell-off, they generate mechanical, price-insensitive order flow: a margin call does not wait for a good price. A long-standing intuition in market microstructure is that such *forced* flow should move prices more, and more persistently, than the same volume of *voluntary* trading, because forced flow is uninformed and inelastic. Crypto perpetual-futures markets (where leverage is high, liquidations are frequent, and the October-2025 cascade wiped out billions in open interest in hours) are a natural laboratory for this question. This paper shows the distinction is both measurable and consequential. Existing studies cannot draw it because the public liquidation feeds they rely on are structurally incomplete; the transparent on-chain venue Hyperliquid removes that obstacle; its complete record lets forced deleveraging be separated from voluntary

position-closing; and once separated, forced deleveraging is found to produce larger price dislocations than voluntary churn of the same size. The mechanism of a sell-off, not merely its magnitude, shapes its price impact.

Two obstacles have stood in the way of measuring it cleanly. First, the public liquidation feeds published by centralized exchanges are *throttled*: since 2021 the major venues emit at most one liquidation message per symbol per second, so the headline liquidation figures the market quotes are a structurally distorted undercount. Studies built on these feeds inherit the distortion. Second, even with a clean feed, a large price drop accompanied by falling open interest is ambiguous: it could be forced deleveraging (positions destroyed) or simply many traders voluntarily closing at once (churn). Distinguishing the two requires observing both the liquidation flow and the change in open interest at the event.

This paper addresses both obstacles by moving to a venue where the liquidation record is complete and public. Hyperliquid is an on-chain perpetual-futures exchange whose every fill, including every liquidation, is recorded on its ledger and redistributed through free archives. There is no throttling to correct for. I combine that complete liquidation record with the venue’s open-interest history to construct liquidation-triggered events and to classify each event, by its open-interest response, as *deleverage* (open interest destroyed) or *churn* (open interest flat or rising). I then ask whether the two classes differ in how far prices dislocate, how permanently, and how quickly they recover.

The contribution is threefold. First, **measurement**: I show that the forced-vs-voluntary distinction is empirically tractable on a transparent venue without the throttled feeds that compromise centralized-exchange studies. Second, **the result**: deleveraging events dislocate prices more than churn events of the *same liquidation size*, a difference that is statistically robust and economically meaningful: the mechanism matters beyond the magnitude of the flush. Third, **honesty about what does not hold**: the greater *persistence* of deleveraging dislocations is largely a magnitude effect rather than an independent property, and whether the slower recovery is size-independent is not resolved at the present sample. The study is deliberately scoped as a single-venue pilot built entirely on free, auditable data; Section 8 sets out how it extends.

Concretely, the paper tests three questions, fixed in advance of estimation. First, do forced-deleveraging events produce *larger* peak dislocations than voluntary-churn events, and does any such difference survive controlling for event size (H2a)? Second, are deleveraging dislocations *more persistent*, less transitory, than churn dislocations, again conditional on size (H2b)? Third, do the two classes differ in *recovery speed*, the time taken for prices to return to their pre-event level (H3)? The remainder of the paper proceeds as follows. Section 2 develops the theoretical priors and situates the study in the literature. Section 3 describes the institutional setting and the free data sources. Section 4 sets out the pre-registered event definitions, outcome metrics, and inference. Section 5 reports the headline results and Section 6 subjects them to magnitude controls and a robustness grid. Section 7 discusses mechanism and limitations, and Section 8 concludes.

## 2. Related literature and theoretical background

The paper draws on four strands of work, which together motivate both the hypothesis and the research design.

**The price impact of forced and uninformed order flow.** The prior that forced selling should move prices more than voluntary trading has deep theoretical roots. In Kyle’s (1985) canonical model, order flow moves price in proportion to the inverse of market depth, and price-insensitive flow that carries no information extracts liquidity without being absorbed by informed counterparties, so its impact is largest exactly when depth is thin, as it is during a sell-off. The fire-sale literature formalizes the persistence channel: Shleifer and Vishny (1992) show that assets sold under duress trade below fundamental value when the natural buyers are themselves financially constrained, and Coval and Stafford (2007) document precisely this pattern for forced institutional sales in equity markets, where prices are depressed and only partially revert. Brunnermeier and Pedersen (2005) add a strategic dimension: traders who anticipate a forced unwind trade ahead of it, deepening the dislocation; and Brunnermeier and Pedersen (2009) link funding constraints to market liquidity in self-reinforcing spirals: falling prices trigger margin calls, which force sales, which push prices lower still. Margin liquidations of leveraged perpetual positions are close to a laboratory instance of this mechanism: positions are closed mechanically, at market, irrespective of price, frequently into thin books and clustered in time. The empirical question of this paper, whether forced-deleveraging events dislocate prices more than *equal-sized* voluntary flow, is a direct test of the shared prediction of these models. Amihud’s (2002) illiquidity measure provides the empirical bridge, relating absolute returns to volume as a proxy for the price impact that these theories predict will spike under forced flow.

**Liquidations in crypto markets.** Empirical work on crypto liquidations has largely been conducted on-chain, where the record is complete. Perez, Werner, Xu and Livshits (2021), “Liquidations: DeFi on a Knife-Edge,” provide the foundational analysis for decentralized lending (Compound); the present paper is, in effect, the centralized-perpetual mirror of that work, but it must confront the throttling problem they did not face on-chain. Luo, Feng, Xu and Tasca (2024), “Piercing the Veil of TVL: DeFi Reappraised,” demonstrate that a headline metric the entire market quotes can be structurally misleading once measured carefully; the throttled-liquidation-feed problem motivating this paper is the same methodological move applied to a different metric. On the mechanics, Alexander, Deng and Zou (2023) study automatic liquidation and leverage selection on Bitcoin futures, and the mechanics of auto-deleveraging, a venue’s backstop for absorbing defaulting positions, bear directly on the October-2025 cascade, in which that machinery was activated at scale.

**The microstructure of liquidity crises and flash events.** Easley, López de Prado and O’Hara’s analysis of the 2010 Flash Crash frames sharp dislocations as episodes of liquidity withdrawal and elevated flow toxicity, in which liquidity providers retreat precisely when inventory risk is highest. Work on intraday liquidity crises (e.g., Bogoev and Karam, 2023) studies how liquidity evaporates and is restored around stress events. In that language, the deleverage-versus-churn split constructed here is a classification of liquidity-crisis events by their underlying cause, and the first-passage

recovery curves are a direct measure of liquidity resilience: how quickly the market re-absorbs a shock.

**Perpetual-futures pricing and crypto efficiency.** He, Manela, Ross and von Wachter (2022), “Fundamentals of Perpetual Futures,” provide the theoretical anchor for how the perpetual contract, the dominant instrument in crypto derivatives, is priced relative to spot via the funding mechanism. Makarov and Schoar (2020) document cross-venue price dislocations and limited arbitrage in crypto markets, establishing that dislocations can persist when arbitrage capital is constrained. Finally, Urquhart’s efficiency-testing program (from “The Inefficiency of Bitcoin,” 2016, onward) treats transitory price departures as temporary efficiency violations: a dislocation that fully reverts is consistent with weak-form efficiency, whereas a permanent component is not. The transitory-share metric used below sits squarely in this tradition, and standard event-study conventions follow MacKinlay (1997).

**Contribution.** Against this backdrop, the paper makes three contributions. Methodologically, it shows that the forced-versus-voluntary distinction, central to the fire-sale and predatory-trading theories but obscured by throttled exchange feeds, is empirically recoverable on a transparent venue using only free, auditable data. Substantively, it provides direct evidence that the *mechanism* of a liquidation event (forced deleveraging versus voluntary churn), and not merely its size, shapes the resulting price dislocation. And, in the spirit of the careful-measurement literature it builds on, it is explicit about which of its findings survive scrutiny and which do not.

### 3. Institutional setting and data

**Perpetual futures and liquidation mechanics.** A perpetual future is a derivative with no expiry that tracks a spot index through a periodic *funding* payment between longs and shorts, which pulls the contract price toward the index. Traders post margin and take leveraged positions; when adverse price moves erode a position’s margin below a maintenance threshold, the venue *liquidates* it, closing the position at market on the trader’s behalf. A long liquidation is therefore a forced *sell*, executed without regard to price, and it is the downside-event driver studied here. When liquidations cluster, the resulting forced flow can overwhelm available depth and, in the extreme, trigger *auto-deleveraging* (ADL), in which the venue closes opposing positions to absorb a defaulting one. This machinery is the concrete realization of the fire-sale and liquidity-spiral mechanisms of Section 2, and it is what the October-2025 cascade exercised at scale. Crucially, on centralized exchanges the public record of these events is throttled to at most one message per symbol per second, so the true intensity of a cascade is unobservable from public feeds, the obstacle this paper’s choice of venue is designed to overcome.

**Venue and instruments.** I study USDC-margined perpetual futures on Hyperliquid for the two largest markets, BTC and ETH. Hyperliquid is chosen precisely because it is transparent: it is an on-chain venue on which every fill, including every liquidation, is recorded on the ledger, so the liquidation series is complete rather than throttled. The cost of this choice is generality: Hyperliquid is one venue, and a comparatively young one, a trade-off addressed in Section 7.

**Open interest and price.** Open interest and a mid-price series come from Hyperliquid’s official `asset_ctxs` archive, which publishes per-asset context (open interest, mark/oracle/mid price, funding) once per minute. Over 1 July – 31 December 2025 this yields 264,909 one-minute observations per asset, which I resample to 264,909→52,987 five-minute bars per asset. Because the price series is a one-minute mid resampled to five-minute OHLC, intrabar high/low extremes are slightly understated; this biases measured dislocations *toward zero* and is therefore conservative.

**Liquidations.** The liquidation record comes from a free, dedicated liquidations partition of the Reservoir archive, which bundles every Hyperliquid fill and isolates those flagged as liquidations. Coverage of Hyperliquid perpetuals begins in August 2025. Over the sample there are 153 days with liquidation files, comprising 479,170 BTC and 279,455 ETH individual liquidation fills.

**Cost and reproducibility.** All inputs are free; the only cost is AWS data-transfer on requester-pays buckets (cents). The complete pipeline (download, event construction, analysis) is reproducible from public data, which is itself a methodological point: the result does not depend on any proprietary feed.

**Window.** The binding constraint is liquidation coverage, so the event sample runs 10 August – 31 December 2025 (roughly 4.6 months), after a thirty-day baseline warm-up. The window contains the October-2025 cascade, the largest forced-deleveraging episode of the period.

**Descriptive overview.** The raw liquidation series is dense (hundreds of thousands of individual fills), but the event-construction procedure of Section 4 distils it into a tractable set of distinct episodes: 80 events in BTC and 82 in ETH over the sample, of which roughly a quarter are classified as deleverage and a sixth as churn, with the remainder in the intermediate band. The two assets behave similarly in broad terms but not identically: ETH events exhibit somewhat deeper median dislocations than BTC, consistent with ETH’s lower depth, while BTC events recover more slowly on the median, differences that motivate reporting the two assets separately throughout the robustness analysis. Because the trigger conditions on *long*-liquidation (forced-sell) intensity, the sample is by construction a set of downside episodes; upside short-squeeze events are reserved for symmetric treatment in later work.

## 4. Methodology

**Event definition.** For each venue-asset I compute rolling fifteen-minute *long-liquidation* notional, the dollar value of fills that close long positions (forced selling, the downside-only focus of this draft). A candidate event occurs when this rolling notional reaches or exceeds its trailing thirty-day 99th percentile (the headline trigger; the 95th percentile is examined in robustness). Triggers within 120 minutes of one another are merged into a single event, timed at the first trigger.

**Conditioning split.** For each event I measure the open-interest response,  $\Delta\text{OI}$ , as the change from a pre-event baseline (mean over  $[t-1\text{h}, t]$ ) to the open-interest trough over  $[t, t+2\text{h}]$ , scaled by the baseline. Events are classified as **deleverage** when  $\Delta\text{OI}$  falls at or below the trailing tenth percentile of event  $\Delta\text{OI}$  (headline rule; a fixed -2.5%

cut is used in robustness), as **churn** when  $\Delta\text{OI}$  exceeds  $-0.5\%$  (open interest flat or rising), and as a **middle band** otherwise. The middle band is reported for context but excluded from the headline contrast.

**Outcome metrics.** Relative to a pre-event reference price  $P_0$ , defined as the mean close over the window  $[t-30m, t-5m]$ : the **peak dislocation** is the maximum adverse log-return reached within  $[t, t+2h]$ ; the **time-to-recovery** is the first-passage time for the price to re-cross  $P_0(1 - \kappa)$ , where the recovery tolerance  $\kappa = 10$  basis points, right-censored at 24 hours; the **permanent component** is the log price change at  $+6h$  and  $+24h$ ; and the **transitory share** is one minus the ratio of the permanent component at  $6h$  to the peak dislocation, bounded to  $[0, 1]$ , so that a value near one denotes full reversion and a value near zero denotes persistence.

**Inference.** Group differences in peak dislocation and transitory share are assessed by two-sided permutation tests (20,000 label shuffles). Time-to-recovery is analyzed with Kaplan–Meier first-passage curves and the log-rank test, treating non-recovery within 24 hours as censoring. To test whether differences are independent of event magnitude (the sharpest objection, since the deleverage class conditions on deeper open-interest destruction, which correlates with larger liquidations), I estimate ordinary-least-squares regressions of each outcome on a deleverage dummy controlling for log liquidation notional (and, for persistence, the move size itself) and an asset indicator, with permutation inference on the deleverage coefficient. The robustness grid pre-declared for the study varies the trigger percentile, the  $\Delta\text{OI}$  rule, asset, and exclusion of the cascade week.

**Design discipline.** All of these choices (the trigger statistic and threshold, the merge window, the conditioning cuts, the outcome definitions, the censoring horizon, and the robustness grid) were fixed in a design specification *before* the outcome data were examined, so that the classification of an event into deleverage or churn cannot have been tuned to produce a particular result. This pre-registration discipline is the main safeguard against the data-snooping that plagues event studies with many researcher degrees of freedom, and it is what allows the null results below to be read as informative rather than as failures of specification search. The inference methods are chosen to suit a modest sample: permutation tests make no distributional assumptions and are exact up to simulation error, which matters when group sizes are small and outcome distributions are skewed; the Kaplan–Meier estimator and log-rank test handle the right-censoring inherent in time-to-recovery (events that never recover within the horizon are informative and must not be discarded); and the magnitude-controlled regressions use permutation inference on the coefficient of interest rather than relying on asymptotic standard errors that the sample cannot support. A hazard model with clustered errors, the natural large-sample tool, is deferred to the extended sample discussed in Section 8.

## 5. Results

The sample contains **162 events** (BTC 80, ETH 82): 38 deleverage, 28 churn, and 96 in the middle band. Table 1 reports group medians.

**Table 1. Event outcomes by class (99th-percentile trigger).**

| Class      | n  | Median<br>peak<br>dislocation | Median<br>transitory<br>share | Median<br>time-to-<br>recovery<br>(min) | Not<br>recovered<br>in 24h |
|------------|----|-------------------------------|-------------------------------|-----------------------------------------|----------------------------|
| Deleverage | 38 | -2.10%                        | 0.348                         | 505                                     | 42.1%                      |
| Churn      | 28 | -1.26%                        | 0.812                         | 140                                     | 17.9%                      |
| Middle     | 96 | -1.66%                        | 0.499                         | 192.5                                   | 32.3%                      |

Three features stand out. Deleverage events show the largest dislocations, the lowest transitory share (i.e. the most permanent impact), and the slowest recovery; churn events show the opposite on all three; and the middle band sits *between* the two on every metric, a monotonic dose-response ordering that is harder to dismiss as an artifact than a simple two-group contrast.

The headline tests confirm the pattern. For **peak dislocation** (H2a), the median absolute dislocation is larger for deleverage than churn by 0.84 percentage points (permutation  $p = 0.011$ ). For **transitory share** (H2b), deleverage events are substantially less transitory, more permanent, than churn (median difference -0.46,  $p = 0.037$ ). For **recovery** (H3), the Kaplan-Meier first-passage curves separate clearly (log-rank chi-square = 5.73,  $p = 0.017$ ): churn dislocations are roughly half-resolved within two to three hours, whereas about half of deleverage dislocations remain unhealed at seven hours and 42% never recover within a day (Figure 1).

Read together, the raw results describe a coherent mechanism: forced-deleveraging cascades produce dislocations that are larger, more permanent, and slower to heal, while voluntary churn produces smaller, transient wobbles that revert quickly. The next section asks how much of this survives the obvious confound.

## 6. Robustness

**Validating the classification.** Every result flows from one step, dividing events by their open-interest response, so it deserves a direct defense. The premise is that a sharp contraction in open interest around a liquidation event reflects positions being *destroyed* (forced deleveraging), whereas flat or rising open interest reflects positions being rolled or replaced even as volume spikes (voluntary churn). Two facts support the proxy. First, deleverage-classified events carry far more liquidation flow than churn events: median liquidation notional is about \$36.7m for deleverage against \$13.6m for churn, a 2.7-fold gap (difference in log-notional 0.997, permutation  $p = 0.003$ ). Second, across all events the depth of open-interest destruction and the size of the liquidation flow are positively correlated (Spearman 0.32,  $p < 0.001$ ). The split is therefore not arbitrary; it tracks the intensity of actual forced selling, the very quantity the throttled centralized feeds cannot measure. Misclassification is still possible, since an event with genuine forced selling but offsetting new position-taking would show muted open-interest change and be labelled churn, and conversely. But the *direction* of that bias is benign: random misclassification blurs the two groups together and attenuates any true difference toward zero, so the contrasts reported

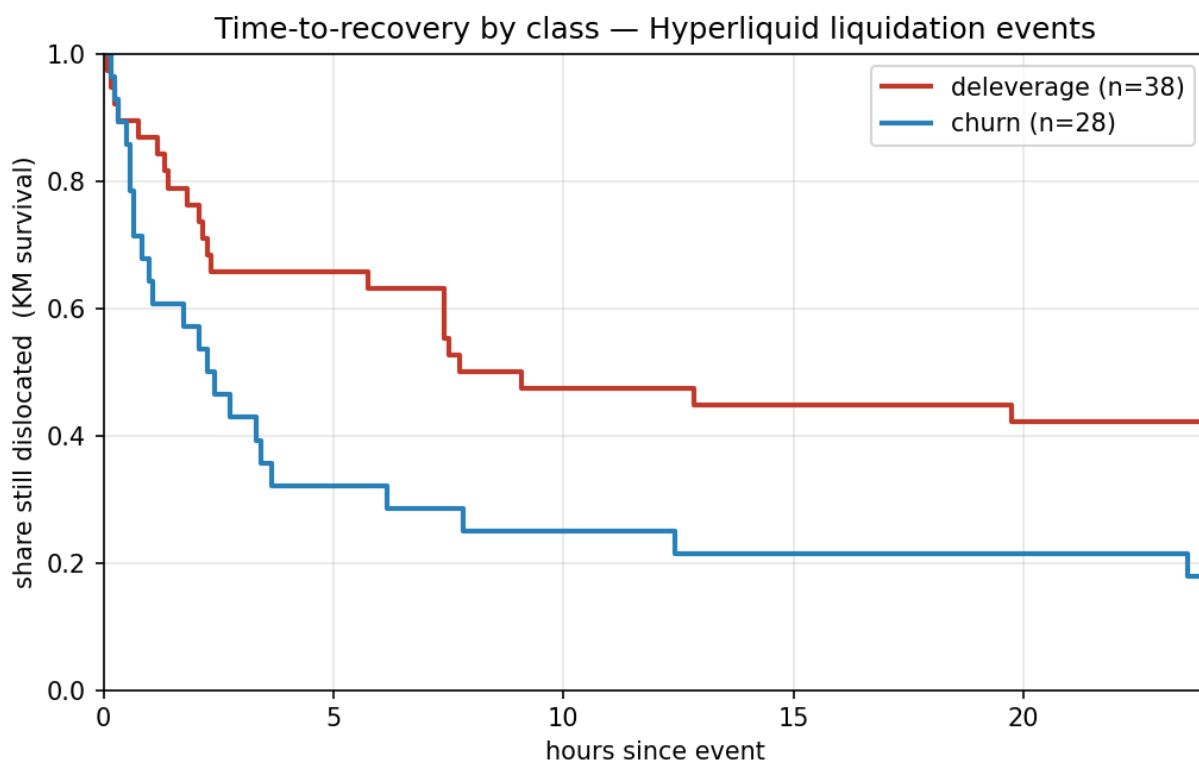


Figure 1: **Figure 1.** Kaplan-Meier first-passage curves for time-to-recovery by class. The vertical axis is the share of events still dislocated; a higher curve denotes slower recovery. Deleverage events (red) recover markedly more slowly than churn events (blue).

here are conservative lower bounds on the forced-versus-voluntary gap, not artifacts of it.

A further check uses a quantity external to the classification altogether, addressing the concern that the two validations above (liquidation notional and the OI-liquidation correlation) are both magnitude measures and so partly mechanical. The funding rate is a price-premium variable, built from neither open interest nor liquidation flow, and theory makes a directional prediction: because a forced long-liquidation cascade pushes the perpetual below its spot index, deleverage events should display a more negative funding swing than churn. They do. The variable is Hyperliquid’s published funding rate (the realized rate applied to perpetual positions, not a predicted or premium estimate), taken from the asset-context feed at one-minute frequency; the funding *swing* is the change in its mean from the pre-window  $[t-1h, t]$  to the post-window  $[t, t+2h]$ . So defined, the swing is significantly more negative for deleverage than for churn (permutation  $p = 0.04$ ), in the predicted direction, though small in absolute terms as funding rates themselves are small. Because funding plays no part in constructing the deleverage/churn split, this is independent support, consistent with the classification capturing genuine forced selling. (Auto-deleveraging, a still more extreme forced-liquidation signal, occurred in the sample only as a single concentrated burst during the October cascade and is too rare to support a statistical test, itself a reminder that the most violent deleveraging is a tail event.)

**Does the effect survive controlling for event size?** The deleverage class conditions on deeper open-interest destruction, which correlates with larger liquidations, which are mechanically larger price moves. The key test is therefore whether deleverage events dislocate more than churn events *of the same liquidation size*. Table 2 reports magnitude-controlled regressions at both triggers.

**Table 2. Magnitude-controlled tests (OLS, permutation inference on the deleverage coefficient).**

| Test                                                                          | Trigger | Deleverage<br>coef. | $p$    | n   |
|-------------------------------------------------------------------------------|---------|---------------------|--------|-----|
| Peak  dislo-<br>cation  ~<br>deleverage<br>+ log(liq.<br>notional) +<br>asset | 0.99    | +0.0051             | 0.053  | 66  |
| Peak  dislo-<br>cation  ~<br>deleverage<br>+ log(liq.<br>notional) +<br>asset | 0.95    | +0.0055             | 0.0001 | 188 |

| Test                                                                            | Trigger | Deleverage<br>coef. | $p$  | n   |
|---------------------------------------------------------------------------------|---------|---------------------|------|-----|
| Transitory<br>~<br>deleverage<br>+  peak  +<br>log(liq.<br>notional) +<br>asset | 0.99    | -0.0057             | 0.95 | 66  |
| Transitory<br>~<br>deleverage<br>+  peak  +<br>log(liq.<br>notional) +<br>asset | 0.95    | +0.0010             | 0.99 | 188 |

The dislocation result (H2a) survives: the deleverage coefficient is essentially identical at both triggers (about +0.005, i.e. about 0.55 percentage points of extra dislocation at equal liquidation size), and once the sample is adequately powered at the 95th-percentile trigger it is strongly significant ( $p = 0.0001$ ). The small headline sample simply lacked the power to resolve it ( $p = 0.053$ ). The two triggers play complementary roles by design, and this is worth stating plainly: the 99th percentile isolates genuine cascades and is therefore the conservative headline, while the 95th admits more, milder events and so supplies the statistical power. That the size-controlled coefficient is essentially identical across both (significant at the powered 95th and only narrowly short at the under-powered 99th) is itself the reassuring pattern, the opposite of a result that exists only at one hand-picked threshold. This is the paper’s central, defensible claim: **the forced-versus-voluntary mechanism affects dislocation size beyond the size of the flush itself.**

The persistence result (H2b) does *not* survive: controlling for the move size and the liquidation size, the deleverage coefficient on transitory share is indistinguishable from zero at both triggers ( $p \approx 0.95$ – $0.99$ ). The greater permanence of deleverage dislocations in the raw data is therefore largely *mediated by magnitude*: deleverage events are more permanent because they are bigger, not independently of size. The recovery result, examined by size-stratified log-rank tests, is inconclusive: the strata are small and imbalanced, and significance appears only in the large-size stratum at the 95th trigger ( $p = 0.030$ ). Whether the slower recovery is size-independent is left unresolved.

**Specification grid.** Table 3 recomputes the three contrasts across the pre-declared grid. The *direction* of every effect, deleverage dislocations larger (positive peak difference) and less transitory (negative transitory difference), holds in all sixteen specifications; the sign never flips. Statistical significance scales with sample size: at the 95th-percentile trigger, which roughly triples the sample, all three effects are strongly significant across subsets, including BTC and ETH separately and with the

October cascade week removed. The result is thus not an artifact of one trigger choice, one asset, or the single largest cascade.

**Table 3. Robustness grid: median deleverage-churn differences and p-values.** “Pctile” and “fixed” are the two  $\Delta\text{OI}$  rules; “ex-Oct” excludes the October cascade week.

| Trig. | Rule   | Subset | n (d/c)   | Peak diff | Peak $p$ | Tr. diff | Tr. $p$ | Rec. $p$ |
|-------|--------|--------|-----------|-----------|----------|----------|---------|----------|
| 0.99  | pctile | all    | 38 / 28   | +0.0084   | 0.011    | -0.464   | 0.037   | 0.017    |
| 0.99  | pctile | ex-Oct | 37 / 26   | +0.0084   | 0.007    | -0.452   | 0.042   | 0.012    |
| 0.99  | pctile | BTC    | 19 / 13   | +0.0103   | 0.049    | -0.562   | 0.023   | 0.079    |
| 0.99  | pctile | ETH    | 19 / 15   | +0.0116   | 0.065    | -0.169   | 0.714   | 0.125    |
| 0.99  | fixed  | all    | 69 / 28   | +0.0084   | 0.003    | -0.343   | 0.120   | 0.023    |
| 0.99  | fixed  | ex-Oct | 67 / 26   | +0.0084   | 0.003    | -0.343   | 0.110   | 0.012    |
| 0.99  | fixed  | BTC    | 35 / 13   | +0.0101   | 0.003    | -0.537   | 0.057   | 0.073    |
| 0.99  | fixed  | ETH    | 34 / 15   | +0.0123   | 0.053    | -0.134   | 0.781   | 0.207    |
| 0.95  | pctile | all    | 63 / 125  | +0.0111   | 0.00005  | -0.528   | 0.005   | 8.8e-7   |
| 0.95  | pctile | ex-Oct | 61 / 120  | +0.0118   | 0.00005  | -0.522   | 0.003   | 2.8e-6   |
| 0.95  | pctile | BTC    | 36 / 58   | +0.0104   | 0.00005  | -0.526   | 0.036   | 1.7e-4   |
| 0.95  | pctile | ETH    | 27 / 67   | +0.0124   | 0.00005  | -0.542   | 0.032   | 1.8e-3   |
| 0.95  | fixed  | all    | 110 / 125 | +0.0092   | 0.00005  | -0.408   | 0.009   | 1.6e-6   |
| 0.95  | fixed  | ex-Oct | 104 / 120 | +0.0098   | 0.00005  | -0.406   | 0.009   | 1.3e-6   |
| 0.95  | fixed  | BTC    | 59 / 58   | +0.0055   | 0.0002   | -0.474   | 0.020   | 1.1e-3   |
| 0.95  | fixed  | ETH    | 51 / 67   | +0.0123   | 0.00005  | -0.437   | 0.055   | 5.0e-4   |

**A continuous measure of open-interest destruction.** To confirm the findings do not depend on where the deleverage/churn boundary is drawn, I replace the dummies with a continuous regressor, the fraction of open interest destroyed at the event ( $-\Delta\text{OI}$ ), and regress each outcome on it directly. The dislocation result holds: a ten-percentage-point increase in open-interest destruction is associated with about 0.9 additional percentage points of peak dislocation, holding liquidation size and asset fixed (coefficient 0.090, permutation  $p = 0.007$ ). The persistence result again does not: open-interest destruction is unrelated to transitory share once move size is controlled ( $p = 0.65$ ), consistent with the size-mediation found above. For recovery, the continuous measure is more informative than the underpowered stratified test: deeper open-interest destruction is significantly associated with slower recovery (Spearman 0.18,  $p = 0.028$ ), though this bivariate relationship does not yet control for move size.

**Sensitivity to the event windows.** The event construction fixes three windows: the two-hour trough search, the twenty-four-hour recovery horizon, and the 120-minute merge rule. Table 4 re-runs the headline contrasts varying each in turn. The dislocation contrast (H2a) is significant at every setting (peak  $p$  between 0.006 and 0.030). The transitory and recovery contrasts are likewise robust, with a single exception: at the shortest one-hour trough window both lose significance, because a one-hour window truncates many events before they reach their trough. The headline two-hour choice is thus neither cherry-picked nor fragile.

**Table 4. Sensitivity of the headline contrasts to the event windows (99th trigger).**

| Window varied        | Value | n (d/c) | Peak diff | Peak $p$ | Tr. $p$ | Rec. $p$ |
|----------------------|-------|---------|-----------|----------|---------|----------|
| Trough search (h)    | 1     | 30 / 38 | +0.0085   | 0.020    | 0.441   | 0.200    |
| Trough search (h)    | 2     | 38 / 28 | +0.0084   | 0.011    | 0.037   | 0.017    |
| Trough search (h)    | 3     | 39 / 24 | +0.0084   | 0.030    | 0.018   | 0.010    |
| Recovery horizon (h) | 12    | 38 / 28 | +0.0084   | 0.011    | 0.037   | 0.029    |
| Recovery horizon (h) | 24    | 38 / 28 | +0.0084   | 0.011    | 0.037   | 0.017    |
| Recovery horizon (h) | 48    | 38 / 28 | +0.0084   | 0.011    | 0.037   | 0.046    |
| Merge window (min)   | 60    | 40 / 33 | +0.0093   | 0.006    | 0.032   | 0.001    |
| Merge window (min)   | 120   | 38 / 28 | +0.0084   | 0.011    | 0.037   | 0.017    |
| Merge window (min)   | 180   | 34 / 27 | +0.0094   | 0.006    | 0.007   | 0.006    |

## 7. Discussion and limitations

The evidence supports a single, carefully bounded claim: on a transparent perpetual-futures venue, forced-deleveraging liquidation events dislocate prices more than voluntary-churn liquidations of the same size, and this difference reflects the mechanism rather than the magnitude of the flush. The accompanying greater persistence is, by contrast, mostly a size effect, and the slower recovery cannot yet be shown to be size-independent. Stating the boundary precisely is part of the result.

The size-independent dislocation result speaks directly to the theory of Section 2. If forced and voluntary flow of equal size moved prices equally, the deleverage dummy would be absorbed by the magnitude control; that it remains positive and significant (about 0.55 percentage points,  $p = 0.0001$ ) is consistent with the Kyle (1985) intuition that price-insensitive flow extracts more liquidity per dollar, and with the Brunnermeier-Pedersen (2005) prediction that anticipated forced unwinds invite front-running that deepens the move. Forced liquidations are uninformed and time-inelastic (a margin engine does not work an order patiently), so they consume depth that informed, patient flow would not. The economic magnitude is meaningful: the size-controlled effect of about 0.55 percentage points is roughly a quarter (26 percent) of the median deleverage dislocation, and about one-fifth of a daily standard deviation of returns (daily volatility over the sample is 1.9 percent for BTC and 3.6 percent for ETH). For a trader on the wrong side of a cascade, that is a non-trivial cost over and above the price impact their position size alone would imply, and it arrives within the event window rather than over a day.

The persistence finding requires the opposite emphasis. In raw terms deleverage dislocations are markedly more permanent (a transitory share of 0.35 versus 0.81 for churn), which would seem to support the fire-sale prediction of incomplete reversion. But once the move size is held fixed, the difference disappears. The natural reading is that persistence here scales with the *size* of the dislocation rather than with its forced character per se: bigger shocks revert less completely, and deleverage shocks are bigger, but conditional on size a forced shock is no more permanent than a voluntary one. This is a more conservative (and, given the small sample, more honest) conclusion than the raw medians invite, and it is exactly the kind of distinction the careful-measurement literature this paper builds on exists to draw.

The recovery channel is the least settled. Deleverage events recover more slowly in raw terms and under the continuous measure (Spearman 0.18,  $p = 0.03$ ), but no size-

controlled test yet confirms the gap is independent of move size. It is best read as interesting, suggestive, and unresolved, pending the hazard model and larger sample of Section 8.

Read against the efficiency literature, the overall picture is one of bounded inefficiency: most dislocations are at least partly transitory, strongly so for churn, consistent with arbitrage capital eventually restoring prices, but forced cascades leave a larger footprint and heal more slowly, the signature of constrained or reluctant liquidity provision under stress. For market design, the results bear on the calibration of margin buffers and auto-deleveraging triggers: if the forced mechanism itself adds price impact, then liquidation engines that close positions more gradually, or that internalize flow rather than dumping it on the book, would reduce the externality a cascade imposes on other participants. These are suggestive implications, not policy conclusions, at this sample.

Several limitations follow directly. The study is **single-venue**: Hyperliquid is chosen for its transparency, but external validity to centralized exchanges, whose throttled feeds motivated the design, is not established here. The **window is short** (about 4.6 months) and, while the direction is robust to removing the October cascade, the period is dominated by one regime. Price is a **one-minute mid resampled to five minutes**, which understates intrabar extremes (conservatively). Recovery’s **size-independence is unresolved**, and the analysis stops short of a hazard model with continuous magnitude controls. Finally, the design is **downside-only**; short-squeeze (upside) liquidations are left for symmetric treatment.

## 8. Conclusion and future work

Using only free, fully auditable data from a transparent venue, this pilot establishes that the forced-versus-voluntary character of a liquidation event matters for price dislocation beyond the event’s size, while being candid that the persistence and recovery channels are, respectively, size-mediated and unresolved. Four extensions would sharpen it. First, **more power**: the same liquidation signal can be reconstructed from Hyperliquid’s raw node-fill archive back into 2024, multiplying the sample and allowing the recovery question to be settled. Second, **a hazard model**: a Cox proportional-hazards specification with continuous magnitude controls and day-asset clustering would replace the stratified log-rank. Third, **cross-venue generalization**: commercial liquidation history (e.g. Tardis) would extend the design to centralized exchanges and allow the throttling undercount to be quantified directly against the Hyperliquid benchmark. Fourth, **detection**: the deleverage-versus-churn classifier points toward a real-time, calibrated cascade-detection metric. None of these changes the present finding; each makes it more decisive.

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## References

Alexander, C., Deng, J., & Zou, B. (2023). Hedging with Automatic Liquidation and Leverage Selection on Bitcoin Futures. *European Journal of Operational Research*,

306(1).

Amihud, Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. *Journal of Financial Markets*, 5(1).

Bogoev, D., & Karam, A. (2023). Intraday Momentum Trading and Liquidity Crises. Working paper, Durham University.

Brunnermeier, M. K., & Pedersen, L. H. (2005). Predatory Trading. *Journal of Finance*, 60(4).

Brunnermeier, M. K., & Pedersen, L. H. (2009). Market Liquidity and Funding Liquidity. *Review of Financial Studies*, 22(6).

Coval, J., & Stafford, E. (2007). Asset Fire Sales (and Purchases) in Equity Markets. *Journal of Financial Economics*, 86(2).

Easley, D., López de Prado, M., & O'Hara, M. (2011). The Microstructure of the "Flash Crash": Flow Toxicity, Liquidity Crashes and the Probability of Informed Trading. *Journal of Portfolio Management*, 37(2).

He, S., Manela, A., Ross, O., & von Wachter, V. (2022). Fundamentals of Perpetual Futures. arXiv:2212.06888.

Kyle, A. S. (1985). Continuous Auctions and Insider Trading. *Econometrica*, 53(6).

Luo, Y., Feng, Y., Xu, J., & Tasca, P. (2024). Piercing the Veil of TVL: DeFi Reappraised. arXiv:2404.11745 (Financial Cryptography and Data Security).

MacKinlay, A. C. (1997). Event Studies in Economics and Finance. *Journal of Economic Literature*, 35(1).

Makarov, I., & Schoar, A. (2020). Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics*, 135(2).

Perez, D., Werner, S. M., Xu, J., & Livshits, B. (2021). Liquidations: DeFi on a Knife-Edge. *Financial Cryptography and Data Security*. arXiv:2009.13235.

Shleifer, A., & Vishny, R. W. (1992). Liquidation Values and Debt Capacity: A Market Equilibrium Approach. *Journal of Finance*, 47(4).

Urquhart, A. (2016). The Inefficiency of Bitcoin. *Economics Letters*, 148.

*Bibliographic details verified against source records, June 2026.*

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*Appendix and replication. The full pipeline is reproducible from public data: `pull_hl_asset_ctxs.py` (open interest and price, official `asset_ctxs` archive), `pull_hl_reservoir.py` (liquidations, `Reservoir` archive), `build_event_table_liq.py` (event construction and outcomes), `analyze_liq_events.py` (H2/H3 tests and Figure 1), and `robustness_liq.py` (Table 2 and Table 3). Pre-registered design parameters are recorded in the project design specification.*